JACC
VOL. 87, NO. 5, 2026
© 2026 BY THE AMERICAN COLLEGE OF CARDIOLOGY FOUNDATION
PUBLISHED BY ELSEVIER

CONCISE CLINICAL GUIDANCE

Transthyretin Cardiac Amyloidosis Evaluation and Management: 2025 ACC Concise Clinical Guidance

A Report of the American College of Cardiology Solution Set Oversight Committee

Writing Committee

Michelle M. Kittleson, MD, PhD, FACC, *Chair*

Amrut V. Ambardekar, MD, FACC
Richard K. Cheng, MD, MSc, FACC
Jan M. Griffin, MD, FACC
Mathew S. Maurer, MD, FACC
Jose Nativi-Nicolau, MD, FACC
Frederick L. Ruberg, MD, FACC

Solution Set Oversight Committee

Gurusher S. Panjrath, MBBS, FACC, *Chair*

Katie Bates, ARNP, DNP, FACC
Eugene H. Chung, MD, MPH, FACC
David M. Dudzinski, MD, JD, FACC
Martha Gulati, MD, MS, FACC
Robert Hendel, MD, MACC
Chayakrit Krittanawong, MD, FACC
Barbara Wiggins, PharmD, FACC
Viet T. Le, PA, FACC, *Ex Officio*

ABSTRACT

Transthyretin amyloid cardiomyopathy has emerged as an increasingly recognized cause of heart failure, particularly in older individuals. There is now greater awareness of transthyretin amyloid cardiomyopathy as an underlying etiology of heart failure, particularly in individuals with musculoskeletal manifestations such as bilateral carpal tunnel syndrome or spinal stenosis. There have also been substantial advances in diagnosis, including the ability to perform accurate noninvasive diagnosis using radionuclide scintigraphy in individuals with a negative monoclonal protein screen. Finally, individuals with transthyretin amyloid cardiomyopathy have benefitted from advances in broadly effective heart failure therapies, namely mineralocorticoid receptor antagonists and sodium glucose-cotransporter 2 inhibitors, as well as specific disease-modifying therapies with transthyretin stabilizers, tafamidis and acoramidis, and the transthyretin silencer vutrisiran. The purpose of this Concise Clinical Guidance is to offer updated strategies to clinicians, reflecting the expanding therapeutic landscape, and reinforcing best practices for the diagnosis and management of transthyretin amyloid cardiomyopathy with a focus on choice of disease-modifying therapies, heart failure therapies, and future directions.

This document was approved by the American College of Cardiology Clinical Policy Approval Committee in October 2025.

The American College of Cardiology requests that this document be cited as follows: Kittleson MM, Ambardekar AV, Cheng RK, Griffin JM, Maurer MS, Nativi-Nicolau J, Ruberg FL. Transthyretin cardiac amyloidosis evaluation and management: 2025 ACC concise clinical guidance. *J Am Coll Cardiol.* 2026;87(5):549-565

Copies: This document is available on the website of the American College of Cardiology (www.acc.org). For copies of this document, please contact Elsevier Inc. Reprint Department via fax (212-633-3820) or e-mail (reprints@elsevier.com).

Permissions: Multiple copies, modification, alteration, enhancement, and/or distribution of this document are not permitted without the express permission of the American College of Cardiology. Requests may be completed online via the Elsevier site (https://www.elsevier.com/about/policies/copyright/permissions).

ISSN 0735-1097/$36.00
https://doi.org/10.1016/j.jacc.2025.09.004

JACC VOL. 87, NO. 5, 2026
FEBRUARY 10, 2026:549–565

TABLE OF CONTENTS

ABSTRACT . . . 549

PREFACE . . . 550

1. INTRODUCTION . . . 551

2. ASSUMPTIONS, DEFINITIONS, AND ABBREVIATIONS . . . 551

2.1. General Clinical Assumptions . . . 551

2.2. Definitions . . . 551

2.3. Abbreviations . . . 552

3. SUMMARY GRAPHIC . . . 552

Figure 1. Challenges in the Evaluation and Management of ATTR-CM . . . 552

4. DESCRIPTION, RATIONALE, AND IMPLICATIONS . . . 553

4.1. Diagnosis . . . 553

4.1.1. Clinical Clues . . . 553

4.1.2. Diagnostic Algorithm . . . 554

Figure 2. Diagnostic Algorithm for Cardiac Amyloidosis . . . 554

4.1.3. Diagnostic Pitfalls . . . 554

4.2. Management . . . 555

4.2.1. Overview of Disease-Modifying Therapies . . . 555

4.2.2. Choosing Among Disease-Modifying Therapies . . . 556

Figure 3. Individual Selection and Choice of ATTR-CM Disease-Modifying Therapies . . . 557

4.2.3. HF Management . . . 558

4.3. Future Therapies . . . 560

4.3.1. Transthyretin Silencing With Gene Editing . . . 560

4.3.2. Transthyretin Depleters With Monoclonal Antibodies . . . 560

4.4. Conclusions . . . 560

REFERENCES . . . 561

APPENDIX 1

Author Relationships With Industry and Other Entities (Relevant) . . . 564

APPENDIX 2

Peer Reviewer Relationships With Industry and Other Entities (Comprehensive) . . . 565

PREFACE

The American College of Cardiology (ACC) has a long history of developing documents (eg, decision pathways, appropriate use criteria) to provide clinicians with guidance on both clinical and nonclinical topics relevant to cardiovascular care. In most circumstances, these documents have been created to complement clinical practice guidelines and to inform clinicians about areas where evidence is new and evolving or where sufficient data are more limited. Despite this, numerous gaps persist, highlighting the need for more streamlined and efficient processes to implement best practices in patient care.

Central to the ACC's strategic plan is the generation of actionable knowledge—a concept that places emphasis on making clinical information easier to consume, share, integrate, and update. To this end, the ACC has shifted from developing isolated documents to creating integrated "solution sets." These are groups of closely related activities, policy, mobile applications, decision-support tools, and other resources necessary to transform care, improve heart health, or both. Solution sets address key questions facing care teams and offer practical guidance to be applied at the point of care. They use both established and emerging methods to disseminate information for cardiovascular conditions and their related management. The success of solution sets rests firmly on their ability to have a measurable impact on the delivery of care. Because solution sets reflect current evidence and ongoing gaps in care, the associated tools will be refined over time to match changing evidence and member needs.

Concise Clinical Guidance (CCG) documents are a key component of solution sets. Highly focused and limited in scope, CCGs provide recommendations where none currently exist, and outline actions required for evidence to be implemented in practice for specific patient populations. CCGs aim to illustrate clinical decision-making processes using tools (ie, figures, tables, checklists) and are limited in scope, focusing on patient populations that share certain characteristics, such as conditions, subtypes, or lines of therapy. In some cases, covered topics will be addressed in subsequent expert consensus decision pathways, appropriate use criteria, clinical practice guidelines, and other related ACC clinical policy as the evidence base evolves. In other cases, these will serve as stand-alone policy and represent best standards.

Gurusher S. Panjrath, MBBS, FACC
Chair, ACC Solution Set Oversight Committee

1. INTRODUCTION

Transthyretin amyloid cardiomyopathy (ATTR-CM) has emerged as an increasingly recognized cause of heart failure (HF) particularly in older individuals. There is now greater awareness of ATTR-CM as an underlying cause of HF[1] and a concomitant diagnosis in aortic stenosis,[2] particularly in individuals with musculoskeletal manifestations such as bilateral carpal tunnel syndrome[3] or spinal stenosis.[4] Substantial advances in diagnosis have been developed, including the ability to perform accurate noninvasive diagnosis using radionuclide scintigraphy in individuals with a negative monoclonal protein screen.[5] Finally, individuals with ATTR-CM have benefitted from advances in effective HF therapies, namely mineralocorticoid receptor antagonists[6-8] and sodium glucose-cotransporter 2 inhibitors,[7-9] as well as disease-modifying therapies with the transthyretin stabilizers tafamidis[10] and acoramidis,[11] and the transthyretin silencer vutrisiran.[12]

The purpose of this CCG is to offer updated strategies to clinicians, reflecting the expanding therapeutic landscape, and reinforcing best practices for the diagnosis and management of ATTR-CM with a focus on choice of disease-modifying therapies, future directions, and unanswered questions.

In accordance with ACC's Relationships With Industry policy, relevant disclosures for the writing committee and comprehensive disclosures for external peer reviewers can be found in Appendices 1 and 2. To ensure complete transparency, a comprehensive listing of relationships with industry and other entities for the writing committee, including relationships not pertinent to this document, is available in an online Supplemental Appendix.

2. ASSUMPTIONS, DEFINITIONS, AND ABBREVIATIONS

2.1. General Clinical Assumptions

1. The principal focus of this effort, including CCG considerations, applies to individuals with or at risk for cardiac amyloidosis.
2. The writing committee endorses the evidence-based approach to HF management recommended in the "2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure."[13]
3. These algorithms assume the managing clinician will seek input as needed from a pharmacist, a cardiologist, a HF specialist, hematologist, an amyloidosis referral center, or a combination of these to guide clinical management.
4. Optimal patient care decisions should properly reflect the individual's preferences and priorities, as well as those of the managing clinician. A shared-decision model regarding care decisions is appropriate particularly when clinical equipoise exists in areas of treatment uncertainty.
5. This CCG is not intended to supersede good clinical judgment. The managing clinician should seek input as needed from relevant experts (eg, pharmacists, cardiologists, endocrinologists).
6. This CCG is based on the best data currently available. New information is being generated rapidly (eg, trials of additional agents and including other patient populations) and, as these data become available, they will impact the considerations made here. Clinicians should be careful to incorporate relevant information published after this CCG is published.
7. Although implementing relevant portions of these recommendations in the acute inpatient setting may be reasonable, this CCG is primarily focused on management in the outpatient ambulatory setting.

2.2. Definitions

Cardiac amyloidosis: a restrictive cardiomyopathy resulting from the deposition of amyloid fibrils in the myocardial interstitium. The 2 most common types are AL amyloidosis, where the amyloid fibrils are composed of monoclonal immunoglobulin light chains, resulting in AL amyloid cardiomyopathy, and ATTR amyloidosis, where the amyloid fibrils are composed of the misfolded transthyretin protein that can occur in the context of variant or wild-type *TTR* genetics, resulting in variant transthyretin amyloid cardiomyopathy (ATTRv-CM) or wild-type transthyretin amyloid cardiomyopathy (ATTRwt-CM), respectively.

- *GDMT:* Guideline-directed medical therapy, representing treatment options supported for use by clinical practice guidelines with specific considerations for use in individuals with ATTR-CM.
- *HF:* Defined per criteria outlined in the "2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure."[13] An HF event, including hospitalization, is defined by the criteria outlined in the "2014 ACC/AHA Key Data Elements and Definitions for Cardiovascular Endpoint Events in Clinical Trials."[14]
- *HF with reduced ejection fraction (HFrEF):* Clinical diagnosis of HF and left ventricular ejection fraction ≤40%.[13]
- *HF with mildly reduced ejection fraction (HFmrEF):* Clinical diagnosis of HF and left ventricular ejection fraction 41% to 49%.[13]
- *HF with preserved ejection fraction (HFpEF):* Clinical diagnosis of HF and left ventricular ejection fraction ≥50%.[13]

JACC VOL. 87, NO. 5, 2026
FEBRUARY 10, 2026:549–565

- *Monoclonal protein screen:* Abnormal serum kappa and lambda free light chain ratio or monoclonal band on serum immunofixation electrophoresis or urine immunofixation electrophoresis which, when taken together, have a >99% sensitivity for the identification of AL amyloidosis.[15]

New York Heart Association (NYHA) functional classification:

- *Class I:* No limitation of physical activity. Ordinary physical activity does not cause symptoms of HF.
- *Class II:* Slight limitation of physical activity. Comfortable at rest, but ordinary physical activity results in symptoms of HF.
- *Class III:* Marked limitation of physical activity. Comfortable at rest, but less-than-ordinary activity causes symptoms of HF.
- *Class IV:* Unable to perform any physical activity without symptoms of HF, or symptoms of HF at rest.

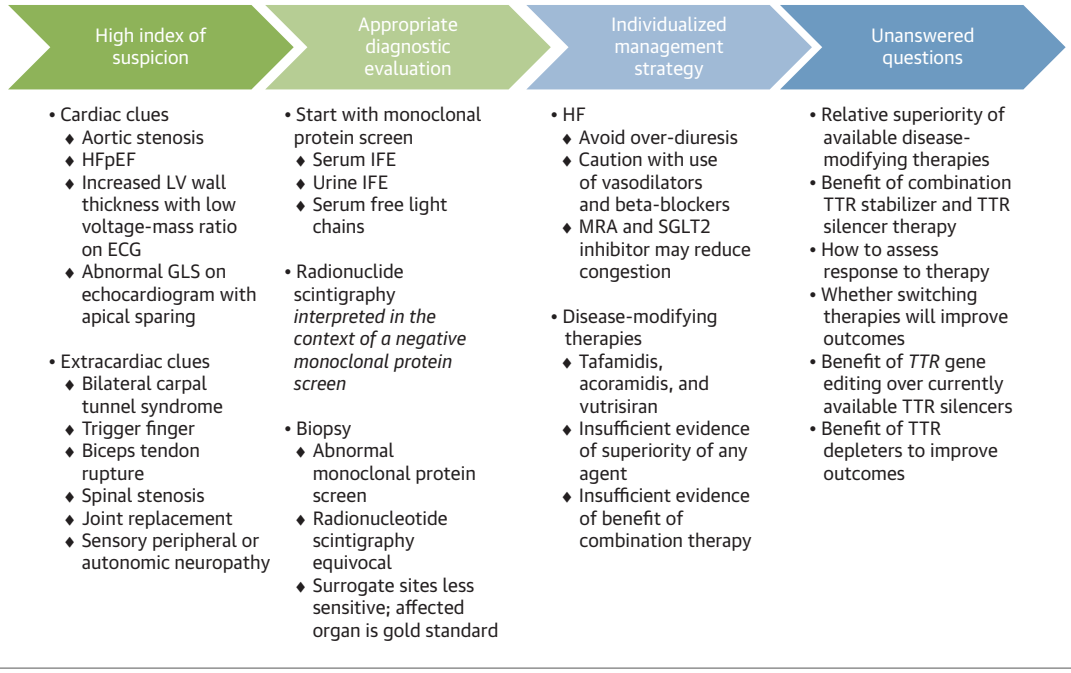
2.3. Abbreviations

ATTR-CM = transthyretin cardiomyopathy
ATTRv-CM = variant transthyretin amyloid cardiomyopathy
ATTRwt-CM = wild-type transthyretin amyloid cardiomyopathy
eGFR = estimated glomerular filtration rate
HF = heart failure
HFmrEF = heart failure with mildly reduced ejection fraction
HFpEF = heart failure with preserved ejection fraction
HFSA = Heart Failure Society of America
MRA = mineralocorticoid antagonist
NYHA = New York Heart Association
SGLT2 = sodium-glucose cotransporter-2

3. SUMMARY GRAPHIC

Figure 1 summarizes the key challenges encountered in the diagnosis, evaluation, and management of ATTR-CM.

FIGURE 1 Challenges in the Evaluation and Management of ATTR-CM



A high index of suspicion is required to diagnose ATTR-CM. Multiple challenges exist in the evaluation and management of ATTR-CM including appropriate diagnostic evaluation, individualized management strategy, and awareness of unanswered questions based on evolving evidence. ECG = electrocardiogram; HFpEF = heart failure with preserved ejection fraction; IFE = immunofixation electrophoresis; GLS = global longitudinal strain; LV = left ventricular; MRA = mineralocorticoid receptor antagonist; SGLT2 = sodium-glucose cotransporter-2; TTR = transthyretin.

4. DESCRIPTION, RATIONALE, AND IMPLICATIONS

4.1. Diagnosis

4.1.1. Clinical Clues

Cardiac amyloidosis is an underrecognized condition and thus requires a high index of suspicion for diagnosis (Table 1). Presentation is heterogenous and nonspecific, often mimicking other more common age-related cardiac conditions. ATTR deposition is observed in about 15% of patients with degenerative aortic valve stenosis[2] or patients with HFpEF.[1,16,17] Extracardiac transthyretin amyloid deposition, typically in soft tissues, often precedes cardiac involvement; about 80% of individuals with ATTR-CM have preceding orthopedic interventions of the shoulder, hip, or knee; carpal tunnel syndrome; spinal stenosis; trigger finger; or biceps tendon rupture.[18] Transthyretin deposition in the ligamentum flavum has been reported in up to one-third of older individuals undergoing surgery for lumbar spinal stenosis.[19,20] Spontaneous biceps tendon rupture may occur a median of 5 years prior to the HF diagnosis,[21] while carpal tunnel syndrome can manifest 5 to 10 years prior to the cardiac diagnosis.[22,23] However, the proportion of individuals with musculoskeletal evidence of transthyretin amyloid deposits that will ultimately develop cardiac manifestations remains incompletely characterized.

Polyneuropathy, particularly peripheral sensory or autonomic dysfunction, may also occur in ATTR. In ATTRv, the presence of peripheral neuropathy varies with the variant. For individuals in the endemic regions of Portugal with the p.Val50Met variant, >80% develop peripheral neuropathy by age 50 years. By contrast, for patients with the p.Val142Ile variant, which is predominantly associated with cardiomyopathy, the prevalence of polyneuropathy is about 10%. ATTRwt, in turn, predominantly causes cardiomyopathy, while approximately 30% of patients may have a polyneuropathy that may (or may not) be attributable to amyloidosis.[24-26]

Electrocardiography, echocardiography, and cardiac magnetic resonance (CMR) imaging (MRI) may suggest the diagnosis (Table 1). Of note, in contrast to AL amyloidosis, low QRS voltage on electrocardiogram (ECG) is not sensitive and only observed in 25% to 40% of cases of ATTR-CM. A more accurate electrocardiographic parameter is voltage-to-mass ratio.[27,28] Echocardiography demonstrates increased left ventricular wall thickness and abnormal global longitudinal strain, often with an apical sparing pattern.[29-31] CMR late gadolinium enhancement imaging and extracellular volume determination provides complementary information,[32] but neither echocardiography nor CMR can definitively diagnose ATTR-CM without tissue biopsy or reliably distinguish ATTR-CM from AL amyloidosis.

TABLE 1 Suggestive Findings for ATTR-CM

Modality	Suggestive Findings
History and physical examination	Cardiac manifestations ■ Fatigue ■ HF symptoms ■ Aortic stenosis ■ Family history of HF ■ Conduction system disease/pacemaker ■ Atrial fibrillation Extracardiac manifestations ■ Bilateral carpal tunnel syndrome ■ Lumbar/cervical spinal stenosis ■ Spontaneous biceps tendon rupture ■ Hip or knee replacement ■ Sensory peripheral neuropathy ■ Family history of neuropathy ■ Autonomic dysfunction
Laboratory test results	■ Persistent low-level troponin elevation ■ Elevated natriuretic peptides
ECG	■ Low voltage to LV mass ratio ■ Atrial fibrillation/flutter especially with controlled ventricular response ■ Conduction delay (atrioventricular block ± intraventricular conduction delay) ■ Pseudoinfarct pattern
Echocardiogram	2-Dimensional imaging ■ Increased LV and RV wall thickness ■ Interatrial septal and diffuse valve thickening ■ RV hypertrophy ■ Pericardial effusion ■ Biatrial enlargement and dysfunction ■ Preserved EF with low stroke volume index Doppler ■ Reduced tissue Doppler s′, e′, and a′ velocities ■ Restrictive transmitral Doppler filling pattern ■ Diastolic dysfunction ≥ grade 2 with high E/A ratio (> 1.5) and reduced E deceleration time (< 150 ms) ■ Reduced TAPSE Other ■ Reduced global longitudinal strain; ratio between EF and global longitudinal strain ■ Apical sparing on 2-dimensional speckle tracking images (classically referred to as "cherry on top") ■ Low flow, low-gradient severe aortic stenosis ■ Impaired left atrial strain
Cardiac MRI	■ Diffuse subendocardial or transmural late gadolinium enhancement ■ Increased myocardial native T1 ■ ECV expansion (>40%) ■ Reversal of nulling (nulling of the myocardium before or at the same inversion time as blood pool)

ATTR-CM = transthyretin cardiac amyloidosis; ECV = extracellular volume; EF = ejection fraction; E/A = early (E) to atrial (A) diastolic flow velocity ratio; LV = left ventricular; MRI = magnetic resonance imaging; RV = right ventricular; TAPSE = tricuspid annular plane systolic excursion.

Cardiac biomarkers may serve as adjunctive tools to exclude ATTR-CM in individuals with a low clinical suspicion; thresholds of N-terminal pro-B-type natriuretic peptide <180 ng/L and high-sensitivity troponin T (hs-TnT) <14 ng/L can exclude ATTR-CM.[33,34]

Although no established evidence-based algorithm exists for translating cardiac and noncardiac clinical clues into a pretest probability of ATTR-CM, a 6-variable score

JACC VOL. 87, NO. 5, 2026
FEBRUARY 10, 2026:549-565

incorporating age, sex, hypertension diagnosis, relative wall thickness, posterior wall thickness, and ejection fraction has been reported and offers insight as a screening tool.[35] Artificial intelligence and machine learning screening algorithms for ECG,[36] echocardiography,[37] and cardiac MRI[38] are emerging tools for the diagnosis of ATTR-CM. Additionally, a summary of high-yield clinical and imaging clues, which should prompt initiation of the diagnostic algorithm, is shown in **Table 1**.

4.1.2. Diagnostic Algorithm

The first and most critical step in the diagnostic evaluation of ATTR-CM is the exclusion of light chain amyloidosis (**Figure 2**). This is achieved by performing serum free light chain analysis, and serum and urine electrophoresis with immunofixation; a combination of these 3 tests has a 99% sensitivity for the detection of AL amyloidosis.[39]

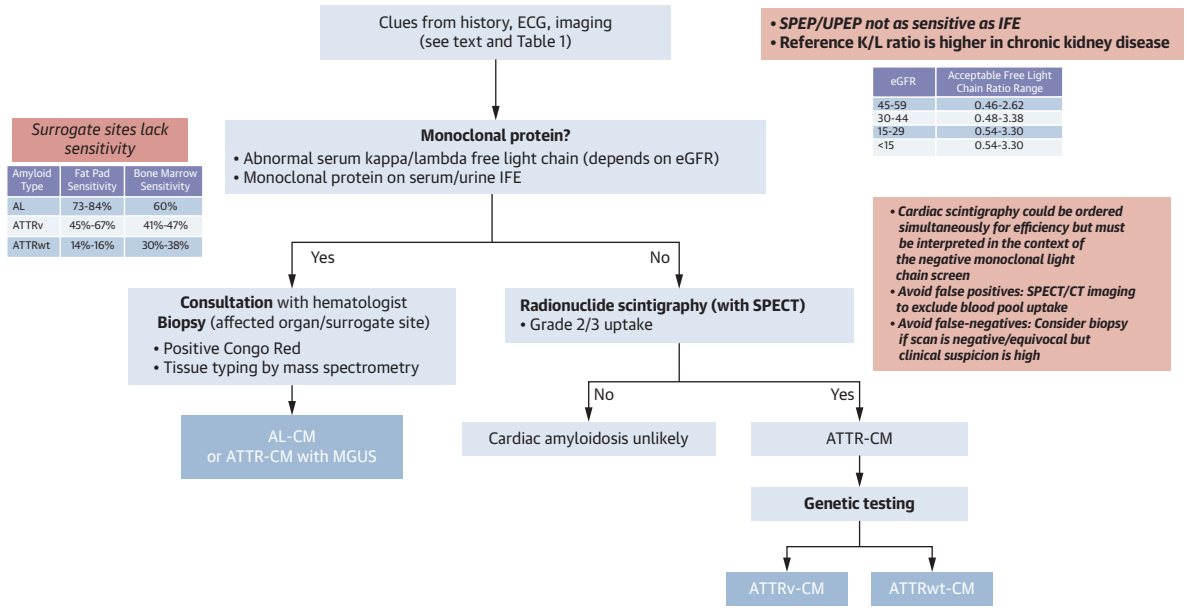
The ratio of kappa and lambda free light chains, rather than the absolute levels, should be used for diagnostic purposes. Of note, the ratio is affected by kidney dysfunction with subsequent reduced renal clearance of kappa light chains resulting in a higher kappa-to-lambda ratio than expected. In such cases, serum and urine immunofixation electrophoresis results are typically normal, and AL amyloidosis is uncommon (**Figure 2**).[40] However, elevated lambda free light chains resulting in a lower kappa-to-lambda ratio should be further investigated. If there is uncertainty regarding interpretation of an abnormal monoclonal protein screen, consultation with a hematologist may be useful to confirm the need for further investigation to confirm the presence of a plasma cell dyscrasia.[41]

Once AL amyloid has been ruled out with a negative monoclonal protein screen, radionuclide scintigraphy can be used for noninvasive diagnosis. Perugini grade 2 or 3 myocardial radiotracer uptake has a 100% positive predictive value for ATTR-CM in the absence of a monoclonal protein.[5] However, any individual with an abnormal monoclonal protein screen result must undergo tissue biopsy and amyloid typing by mass spectrometry for definitive diagnosis.

4.1.3. Diagnostic Pitfalls

Care must be taken to avoid the common diagnostic pitfalls. In individuals with suspected cardiac amyloidosis

FIGURE 2 Diagnostic Algorithm for Cardiac Amyloidosis



AL-CM = amyloid monoclonal immunoglobulin light chain cardiomyopathy; ATTR-CM = transthyretin amyloid cardiomyopathy; ATTRv-CM = variant transthyretin amyloid cardiomyopathy; ATTRwt-CM = wild-type transthyretin amyloid cardiomyopathy; CMR = cardiac magnetic resonance; CT = computed tomography; ECG = electrocardiogram; eGFR = estimated glomerular filtration rate; IFE = immunofixation electrophoresis; K/L = kappa/lambda; MGUS = monoclonal gammopathy of undetermined significance; SPECT = single-photon emission computed tomography; SPEP/UPEP = serum/urine protein electrophoresis. Range of sensitivity of surrogate site biopsies adapted from Cohen et al[42]; Quarta et al[43]; Fine et al[44]; Gertz et al[45]; Kyle et al[46]; Hansen et al[47]; and van Gameren et al.[48] Range of normal K/L ratio by eGFR from Long et al[40] and Hutchison et al.[49]

and an abnormal monoclonal protein screen, biopsy of a surrogate site (eg, the abdominal fat pad or bone marrow) offers low sensitivity (Figure 2).[42-44] Thus, although helpful if positive, a negative fat pad biopsy result excludes neither AL amyloid cardiomyopathy nor ATTR-CM, and biopsy of an affected organ may be required to adjudicate the diagnosis.

For the nonbiopsy diagnosis of ATTR-CM, strict adherence to standardized radionuclide imaging protocols is essential.[29] ^{99m}Technetium (^{99m}Tc) bone-avid compounds (including pyrophosphate), DPD (3,3-diphosphono-1,2-propanodicarboxylic acid), and HDP (hydroxymethylene diphosphonate) are used. Pyrophosphate is historically used in the United States, although HDP can be used in the case of pyrophosphate shortages to avoid diagnostic delays or as an alternative imaging agent.[50] Diffuse myocardial uptake on planar imaging must be differentiated from blood pool activity using single-photon emission computed tomography or single-photon emission computed tomography/computed tomographic imaging techniques, at 2 to 3 hours after radiotracer injection to avoid misclassification.[51] Other causes of a false-positive scintigraphy scan include prior myocardial infarction, rib fractures overlying the heart, hydroxychloroquine toxicity, and AL amyloidosis. Conversely, false-negative scans can occur in individuals with certain ATTR variants (p.Phe84Leu and p.Val50Met) or in the early stages of disease.

4.2. Management

4.2.1. Overview of Disease-Modifying Therapies

4.2.1.1. Transthyretin Stabilizers

Transthyretin stabilizers preserve the normal tetrameric structure of transthyretin to prevent dissociation, misfolding, and aggregation into amyloid fibrils that deposit into tissues. In 2019, tafamidis became the first U.S. Food and Drug Administration (FDA)-approved transthyretin stabilizer for ATTR-CM based on the results of ATTR-ACT (Tafamidis in Transthyretin Cardiomyopathy Clinical Trial), which randomized 441 individuals with ATTR-CM to tafamidis versus placebo.[10] After 30 months, tafamidis demonstrated a lower all-cause death and frequency of cardiovascular-related hospitalizations (Table 2). A long-term extension study with a median follow-up of 51 months showed a 30% relative reduction in the risk of death with 80 mg compared with 20 mg.[52] Tafamidis 61 mg free acid is bioequivalent to tafamidis meglumine 80 mg (4 × 20 mg),[53] both of which are FDA-approved for ATTR-CM. Although well-tolerated, caution should be used when administering tafamidis with rosuvastatin

TABLE 2 Overview of FDA-Approved Disease-Modifying Therapies for ATTR-CM

	Tafamidis	Acoramidis	Vutrisiran
Mechanism of action	Transthyretin stabilizer at T4 binding site	Transthyretin stabilizer hydrogen bonds at T4 binding site (mimicking the stabilizing transthyretin p. Thr139Met variant)	siRNA *TTR* gene silencer
Trial	ATTR-ACT[10]	ATTRibute-CM[11]	HELIOS-B[12]
Years conducted (published)	2013-2018 (2018)	2019-2023 (2024)	2019-2024 (2025)
Study type	Randomized, double-blind, placebo-controlled phase 3	Randomized, double-blind, placebo-controlled phase 3	Randomized, double-blind, placebo-controlled phase 3
Duration, mo	30	30	33-36
Study dose	20 mg or 80 mg orally once a day; results pooled in primary analysis	800 mg orally twice a day	25 mg subcutaneous every 3 mo
FDA-approved dose	Tafamidis 61 mg or tafamidis meglumine 80 mg	712 mg twice daily	25 mg subcutaneous every 3 mo delivered in a healthcare setting
No. of participants	441	632	654
Mean age, y	74	77	77
Male sex, No. (%)	398 (90)	570 (90)	605 (93)
Race, No. (%)			
White	357 (81)	555 (88)	552 (84)
Black	63 (14)	30 (5)	47 (7)
Asian	18 (4)	13 (2)	37 (6)
TTR genotype, No. (%)			
ATTRwt	335 (76)	571 (90)	578 (88)
ATTRv	106 (24)	61 (10)	76 (12)
NYHA class, No. (%)			
Class I	37 (8)	68 (11)	84 (13)
Class II	263 (60)	455 (72)	508 (78)
Class III	141 (32)	109 (17)	62 (9)
NT-proBNP, pg/mL, median	3078	2872	1911
6MWD, m, mean	351.9	354.8	374.5
KCCQ-OSS, mean	66.6	70.9	72.65
NAC stage, No. (%)			
1	190 (42)	361 (57)	437 (67)
2	167 (38)	203 (32)	187 (28)
3	84 (20)	68 (11)	30 (5)
Tafamidis use, No. (%)			
Treatment group	264 of 264 (100)	61 of 409 (15)*	133† and 44‡ of 326 (53)
Placebo group	0 of 177 (0)	46 of 202 (23)*	129† and 41‡ of 328 (52)

Continued on the next page

TABLE 2 Continued

	Tafamidis	Acoramidis	Vutrisiran
Primary outcome	Death, CV hospitalizations win ratio: 1.7 (95% CI, 1.3-2.3)	Death, CV hospitalizations, NT-proBNP change, 6MWT change win ratio 1.8 (95% CI, 1.4-2.2)	Death, recurrent CV events: HR: 0.72 (95% CI: 0.56-0.93)
All-cause death, HR (95% CI)	0.70 (0.51-0.96)§	0.77 (0.54-1.1)§	0.69 (0.49-0.98)§
All-cause death, %, treatment vs placebo	29.5 vs 42.9§	19.3 vs 25.7§	16 vs 21§
CV Hospitalizations, RRR (95% CI)	0.68 (0.56-0.81)	0.496 (0.36-0.70)‖	0.73 (0.61-0.88)‖
6MWD, LS mean change (95% CI)	75.7 (57.6-93.8)	39.6 (21.1-58.2)	26.5 (13.4-39.6)
KCCQ-OSS, LS mean change (95% CI)	13.65 (9.5-17.8)	9.94 (6.0-13.9)	5.8 (2.4-9.2)
Side effects¶	Similar to placebo	Diarrhea (12%) Abdominal pain (5%)	Pain in extremity (15%) Arthralgia (11%) Dyspnea (7%) Vitamin A decreased (7%) Injection site reaction (4%)
Vitamin A supplementation	Not required	Not required	Supplement containing the locally RDA of vitamin A
Annual cost#	$267,987	$244,539	$477,404

*Tafamidis only permitted after 12 mo.
†Tafamidis use at baseline.
‡Tafamidis use initiated after randomization.
§During double-blind placebo-controlled period: 30 mo for tafamidis and acoramidis and 36 mo for vutrisiran.
‖Recurrent cardiovascular events that included urgent visits for heart failure (adjudicated by a Clinical Events Committee in ATTRibute-CM and nonadjudicated in HELIOS-B).
¶Reported in the FDA labels.
#Average wholesale price per year from Micromedex Red Book as of July 11, 2025.[58]

6MWD = 6-minute walk distance; ATTR-ACT = Tafamidis in Transthyretin Cardiomyopathy Clinical Trial; ATTRibute-CM = Efficacy and Safety of AG10 in Subjects With Transthyretin Amyloid Cardiomyopathy; ATTR-CM = transthyretin amyloid cardiomyopathy; ATTRv = variant transthyretin amyloidosis; ATTRwt = wild-type transthyretin amyloidosis; CV = cardiovascular; FDA = U.S. Food and Drug Administration; HELIOS-B = A Study to Evaluate Vutrisiran in Patients With Transthyretin Amyloidosis With Cardiomyopathy; HR = hazard ratio; KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire - Overall Summary Score; LS = least squares; NAC = National Amyloidosis Centre; NT-proBNP = N-terminal pro-B-type natriuretic peptide; NYHA = New York Heart Association; RDA = recommended dietary allowance; RRR = relative risk reduction; siRNA = small interfering ribonucleic acid; TTR = transthyretin.

>20 mg or atorvastatin >40 mg due to the risk of rhabdomyolysis.

Acoramidis is another transthyretin stabilizer[54] and was FDA-approved in 2024 after the results of the ATTRibute-CM (Efficacy and Safety of AG10 in Subjects With Transthyretin Amyloid Cardiomyopathy) clinical trial that compared acoramidis hydrochloride 800 mg twice a day to placebo in 632 individuals.[11] After 30 months, acoramidis was associated with a favorable effect on all-cause death and cardiovascular hospitalizations[11,55,56] (Table 2). The FDA-approved dose is 712 mg (equivalent to acoramidis hydrochloride 800 mg) orally twice a day.

4.2.1.2. Transthyretin Silencers

Transthyretin silencers cause degradation of transthyretin messenger ribonucleic acid in hepatocytes, reducing transthyretin production.[57] The 2 types of gene silencer approaches studied for amyloidosis include small-interfering ribonucleic acids and antisense oligonucleotides .

Vutrisiran was approved by the FDA in 2022 for ATTRv with polyneuropathy; approval for ATTR-CM occurred in 2025. The HELIOS-B (A Study to Evaluate Vutrisiran in Patients With Transthyretin Amyloidosis With Cardiomyopathy) trial compared vutrisiran 25 mg subcutaneous every 12 weeks versus placebo in 655 individuals.[12] Vutrisiran was associated with a lower risk of death from any cause and recurrent cardiovascular events (Table 2). Because transthyretin is the major transporter of retinol, vitamin A supplementation is advised for any therapy that reduces production of circulating transthyretin. The FDA-approved dose of vutrisiran is 25 mg subcutaneous every 3 months with vitamin A supplementation.

4.2.2. Choosing Among Disease-Modifying Therapies

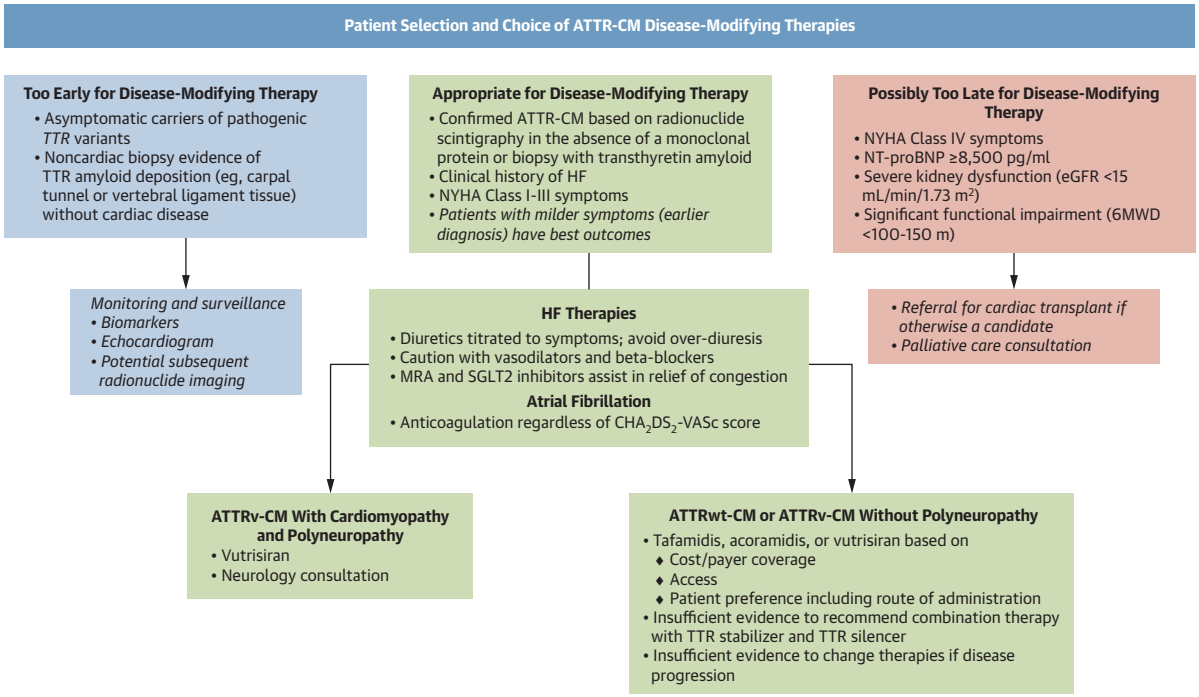
4.2.2.1. Selecting Appropriate Individuals for ATTR-CM Disease-Modifying Therapies

By their mechanisms of action, the transthyretin stabilizers tafamidis and acoramidis and the transthyretin silencer vutrisiran all slow progression, but do not reverse end-organ damage, and thus should be expected to preserve, but not improve, functional status and quality of life. The optimal time to start ATTR-CM disease-modifying therapy is at the earliest onset of symptomatic disease before significant end-organ dysfunction has ensued (Figure 3).

See Kittleson et al[41] for details on screening recommendations for asymptomatic carriers of pathogenic TTR variants or those with noncardiac transthyretin amyloidosis with no evidence of cardiac involvement, as well as for details on management of arrhythmias in ATTR-CM.

Generally, appropriate individuals for ATTR-CM disease-modifying therapy reflect landmark clinical trial criteria: confirmed diagnosis of ATTR-CM and symptomatic HF with NYHA class I-III symptoms, including those with hospitalization for HF or use of loop diuretic, and a left ventricular wall thickness of ≥12 mm.[10-12] However, these criteria should be applied with nuance, because: 1) earlier diagnosis may obviate hospitalization; and 2)

FIGURE 3 Individual Selection and Choice of ATTR-CM Disease-Modifying Therapies



6MWD = 6-minute walk distance; ATTR-CM =transthyretin cardiac amyloidosis; ATTRv-CM = variant transthyretin amyloid cardiomyopathy; ATTRwt-CM = wild-type transthyretin cardiac amyloidosis; eGFR = estimated glomerular filtration rate; HF = heart failure MRA = mineralocorticoid antagonist; NT-proBNP = N-terminal pro-B-type natriuretic peptide; NYHA = New York Heart Association; SGLT2 = sodium-glucose cotransporter-2; TTR = transthyretin.

improvements in HF management with sodium-glucose cotransporter-2 (SGLT2) inhibitors and mineralocorticoid receptor antagonists (MRAs) may control volume status without need for a loop diuretic. Thus, the lack of HF hospitalizations or need for loop diuretics should not preclude prescription of ATTR-CM disease-modifying therapy. Furthermore, because some individuals with ATTR-CM (particularly women)[59] may not have a wall thickness of ≥12 mm, absence of this specific criterion should not preclude initiation of therapy.

Notably, the cohorts for which data do not exist regarding potential benefit from disease-modifying therapy, include the following scenarios: 1) cascade screening of family members of individuals with ATTRv-CM will identify asymptomatic carriers of pathogenic TTR variants; and 2) individuals with carpal tunnel syndrome and spinal stenosis may have transthyretin amyloid involvement of ligamentous structures but no evidence of cardiac disease. Although these individuals have a greater chance of subsequent ATTR-CM development,[3,4,19,22] current evidence does not support therapy initiation. However, ACT-EARLY (Acoramidis Transthyretin Amyloidosis Prevention Trial in the Young Study in Asymptomatic Carriers of a Pathogenic TTR Variant; NCT06563895) will assess the role of the transthyretin stabilizer acoramidis on the development of cardiomyopathy and polyneuropathy. Also, the nonsteroidal anti-inflammatory drug diflunisal can stabilize the transthyretin tetramer and slow progression of ATTRv polyneuropathy,[60,61] and prophylactic use warrants further investigation.

Some individuals with disease may be too advanced to derive benefit from disease-modifying therapy. All 3 trials that resulted in drug approval excluded individuals with advanced symptoms. Specifically, exclusion criteria were: 1) significant functional impairment (6-minute walk distance <100 m for ATTR-ACT and <150 m for ATTRibute-CM and HELIOS-B); 2) high natriuretic peptide levels (N-terminal pro-B-type natriuretic peptide >8,500 pg/mL for ATTRibute-CM and HELIOS-B); and 3) severe kidney impairment (estimated glomerular filtration rate [eGFR] <15 mL/min/1.73 m^2 in ATTRibute-CM; eGFR <25 mL/min/1.73 m^2 in ATTR-ACT; and eGFR <30 mL/min/1.73 m^2 in HELIOS-B).[10-12] Although therapy may offer no harm in individuals with advanced ATTR-CM, benefits have not been demonstrated. For such individuals, referral for heart transplantation evaluation if otherwise eligible based

on comorbidities, or palliative care consultation if not, would be appropriate.[62]

4.2.2.2. Choosing Between Stabilizer Versus Silencer Disease-Modifying Therapy

Because all 3 landmark trials of tafamidis, acoramidis, and vutrisiran were placebo-controlled, no direct randomized clinical trial or comparative effectiveness evidence is available to offer insight into the relative superiority of any agent relative to another. Because all have minimal side effects, this consideration would also not affect choice of therapy. In addition, comparisons of individual clinical trials are limited by differences in trial populations; more contemporary trials enrolled individuals at earlier disease stages as evidenced by lower N-terminal pro-B-type natriuretic peptide, higher 6-minute walk distance, better quality of life, and successively lower event rates even in the placebo arms of each trial (**Table 2**). Additionally, no evidence is available to date that differential changes in serum transthyretin levels after stabilizer therapy are indicative of clinical superiority. Thus, without active controlled clinical trials, it remains unknown whether tafamidis, acoramidis, or vutrisiran is the superior initial therapy for individuals with ATTR-CM.

Consequently, based on current data, the choice of therapy should be based on factors including: 1) cost; 2) access based on payor preferences; and 3) individual preference regarding route and frequency of administration (oral once per day versus oral twice per day versus subcutaneous injection every 3 months with vitamin A supplementation).

The notable exception is an individual with ATTRv-CM and amyloid polyneuropathy. In such individuals, transthyretin silencer therapy with vutrisiran may be preferred because clinical trials show benefit of transthyretin silencers in ATTRv-polyneuropathy.[63-65] Of note, the polyneuropathy observed in individuals with ATTRv-CM is less common than the more common musculoskeletal manifestations (eg, carpal tunnel syndrome, spinal stenosis).[25,26] Although transthyretin silencer therapy is effective in ATTRv-polyneuropathy, no data suggest that transthyretin silencer therapy mitigates the musculoskeletal manifestations. Therefore, transthyretin silencer therapy is not indicated specifically to address these conditions.

4.2.2.3. The Role of Combination Stabilizer and Silencer Therapy

In ATTR-CM, the rationale for combination therapy is based on the potential to enhance efficacy and reduce the risk of disease progression, however at the expense of increased cost (with a lower financial impact when generic options become available). To date, an additive effect is unknown of stabilizing remaining circulating transthyretin after significant transthyretin knockdown.

Current clinical trial data do not justify the use of combination therapy with either 2 stabilizers or a combination of a stabilizer and silencer. In HELIOS-B, 40% of enrolled individuals were on tafamidis at baseline. The remaining 60% of patients comprised the monotherapy group, in whom another 20% initiated tafamidis after randomization at a median of about 17 months. Although the HELIOS-B study was not specifically powered to address the question of combination therapy of vutrisiran and tafamidis, addition of vutrisiran to baseline tafamidis therapy did not result in statistically significant improvements in clinical endpoints in this subgroup.[66-68] Thus, routine combination transthyretin silencer and stabilizer therapy is not recommended based on current available evidence.

The ongoing CARDIO-TTRansform (Study to Evaluate the Efficacy and Safety of Eplontersen in Participants with Transthyretin-Mediated Amyloid Cardiomyopathy; NCT0413617) trial may clarify this issue. CARDIO-TTRansform has enrolled 1,432 participants, 70% of whom were receiving concomitant tafamidis,[69] with results expected in 2026. Given the high number of participants on concurrent stabilizer therapy, the trial may provide critical information about combination therapy in this population, although the power to detect a benefit of combination versus single therapy is still limited.

4.2.2.4. Progression of Disease and Role of Adjustments to Therapy

Currently, no widely accepted definition is available of what constitutes response to or failure of disease-modifying therapy. Although many individuals with ATTR-CM develop worsening HF symptoms, new arrhythmias, or both, it is unclear if this represents sequela from baseline or ongoing ATTR fibril deposition.[70] The advanced age of the affected population (median age, 75-77 years in the ATTR-CM clinical trials)[10-12] and comorbid conditions (eg, kidney dysfunction, frailty) may account for disease progression more than mechanisms linked to transthyretin misfolding and aggregation. At this time, evidence is insufficient to guide adjustment in disease-modifying therapy for worsening symptoms.

4.2.3. HF Management

4.2.3.1. Effects of Restrictive Physiology

Individuals with restrictive physiology exhibit physiologic alterations in the pressure-volume relationships with impaired ventricular compliance leading to elevated left- and right-sided filling pressures, dependence on heart rate to maintain cardiac output due to a decreased

and relatively fixed stroke volume, and impaired vascular-ventricular coupling due to increased ventricular stiffness and abnormal arterial elastance.[41,71] These hemodynamic changes result in preload dependence and afterload intolerance, requiring a tailored approach to HF management (Figure 3, Table 3).

4.2.3.2. Diuretics

Loop diuretics alleviate pulmonary and systemic congestion but should be titrated cautiously because of preload dependence and risk for hypotension, reduced stroke volume, and acute kidney injury from overdiuresis. In individuals with diuretic resistance, switching from furosemide to the more bioavailable torsemide or bumetanide, or using sequential nephron blockade by adding thiazides (eg, metolazone), can be considered with careful monitoring of electrolytes. Higher baseline diuretic requirements[72] and the need for diuretic intensification[67,73] are associated with a worse prognosis.

4.2.3.3. Indications and Pitfalls of Traditional Guideline-Directed Medical Therapy

Historically, the use of beta-blockers in individuals with ATTR-CM raised concern because of dependence on heart rate to maintain cardiac output, chronotropic incompetence from amyloid infiltration of the conduction system, and autonomic dysfunction. Observational data remain mixed, and beta-blocker discontinuation among studies was high, ranging from 22% to 43%.[7,74,75] One study reporting no benefit included individuals with more advanced disease and used higher doses of nonselective beta-blockers.[74] By contrast, the largest cohort suggested that low-dose beta-blockers (bisoprolol ≤2.5 mg/d) may be beneficial in individuals with an ejection fraction ≤40%.[7] Thus, beta-blockers may be cautiously considered at low doses in those with an ejection fraction ≤40%, especially if needed for rate control of atrial fibrillation, with a low threshold for reduction or discontinuation for intolerance.

Retrospective observational studies suggest no clinical benefit with vasodilating angiotensin-converting enzyme inhibitors or angiotensin receptor blockers regardless of stratification by reduced ejection fraction and with high discontinuation rates of 33% to 59% due to side effects.[7] However, a post-hoc analysis of PARADIGM-HF (Prospective Comparison of ARNI [Angiotensin Receptor-Neprilysin Inhibitor] With ACEI [Angiotensin Converting-Enzyme Inhibitor] to Determine Impact on Global Mortality and Morbidity in Heart Failure), demonstrated benefit of sacubitril and valsartan in a subpopulation of individuals with HFmrEF and HFpEF who may have had amyloidosis identified with an ATTR-CM clinical

TABLE 3 Considerations for Guideline-Directed Medical Therapy in ATTR-CM

HF GDMT Class	Potential Indications in ATTR-CM	Concerns for Use in ATTR-CM	Specific Considerations for HF and ATTR-CM
Beta-blockers	■ LV dysfunction ■ Atrial fibrillation with rapid ventricular response ■ Coexisting coronary artery disease ■ Ventricular arrhythmias	■ Low stroke volume and dependence on HR to augment cardiac output ■ Chronotropic incompetence from conduction disease ■ Orthostatic hypotension from autonomic dysfunction	■ No randomized clinical trials to date ■ Observational data with mixed results regarding benefit, likely due to differences in baseline characteristics[7,74,75] ■ High discontinuation rates of 22%-43% in retrospective studies[7,74,75] ■ Consider low-dose beta-blocker in individuals with EF <40% or other indications, such as AF rate control ■ Consider selective beta-1 beta-blocker (eg, metoprolol, bisoprolol) to mitigate hypotension ■ Reduce or discontinue if symptomatic
ACEI/ARB/ARNI	■ LV dysfunction ■ Systemic hypertension	■ Intolerance of afterload reduction due to fixed stroke volume and impaired ventricular-vascular coupling, resulting in hypotension	■ No randomized clinical trials to date ■ Observational data has not shown benefit with ACEI/ARB regardless of reduced EF[7,74] ■ High discontinuation rates of ACEI/ARB of 33%-59% in retrospective studies ■ Retrospective data suggests ARNI may have benefit[8]
Mineralocorticoid receptor antagonists	■ LV dysfunction ■ Augments diuresis and potassium-sparing diuretic	■ Preload dependence	■ No randomized clinical trials to date ■ Retrospective data have suggested benefit with spironolactone[6-8] ■ Well-tolerated ■ Prescribe to relieve congestion and for potassium-sparing properties
SGLT2 inhibitors	■ LV dysfunction ■ Augments diuresis	■ Preload dependence	■ No randomized clinical trials to date ■ Retrospective data have suggested benefit of reduction in diuretic use[8,9,76] ■ Generally well-tolerated ■ Prescribe to relieve congestion and limit loop diuretic use

ACEI = angiotensin-converting enzyme inhibitor, AF = atrial fibrillation; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; ATTR-CM = transthyretin cardiac amyloidosis; EF = ejection fraction; GDMT = guideline-directed medical therapy; HF = heart failure; HR = heart rate; LV = left ventricular; SGLT2 = sodium-glucose cotransporter-2.

score.[35] Although vasodilators may not be tolerated due to presence of symptomatic orthostasis or acute kidney injury, angiotensin receptor neprilysin inhibitor may be cautiously attempted based on these data.

Similarly, a subgroup analysis of TOPCAT-Americas (Treatment of Preserved Cardiac Function Heart Failure With Aldosterone Antagonist in the Americas) of individuals with HFmrEF and HFpEF randomized to spironolactone versus placebo and similarly enriched for amyloidosis[35] demonstrated comparable treatment benefit to the overall population.[8] Comparable findings have been observed in other retrospective analyses[6] with acceptable tolerance with a discontinuation rate of only 8%.[7] No evidence to date has been observed with eplerenone or finerenone in ATTR-CM, although the benefit of MRAs could be considered likely a class effect. MRAs are an option to relieve congestion in individuals with ATTR-CM requiring diuretics.

SGLT2 inhibitors may also provide clinical benefit in individuals with ATTR-CM.[9,76] In a retrospective analysis of the DELIVER trial (Dapagliflozin Evaluation to Improve the Lives of Patients With Preserved Ejection Fraction Heart Failure), benefit to dapagliflozin was observed in individuals with HFmrEF and HFpEF with a phenotype consistent with ATTR-CM.[8] In a multicenter cohort study, SGLT2 inhibitors had a discontinuation rate of only 4.5% and associated with improvements in functional status, reductions in cardiac biomarkers, stabilization of kidney function, and a reduction in all-cause death and HF hospitalization.[9] Supporting these findings, a single-center cohort showed that SGLT2 inhibitors use was associated with greater weight reductions and lower loop diuretic dosage.[76] Taken together, this suggests that both MRA and SGLT2 inhibitors are options to effectively relieve congestion in individuals with ATTR-CM requiring diuretic therapy. However, the routine use of MRA and SGLT2 inhibitors in all individuals with ATTR-CM is not yet supported by evidence.

4.3. Future Therapies

4.3.1. Transthyretin Silencing With Gene Editing

The demonstrated efficacy of transthyretin messenger ribonucleic acid silencers suggests that permanent modification of genomic *TTR* would be similarly effective. Preliminary data regarding the CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) trial-Cas9 agent nexiguran ziclumeran (nex-z) demonstrate durable reduction in transthyretin levels and stabilization of cardiac biomarkers and functional capacity at 12 months.[69] The primary advantage of nex-z is single-dose administration, as well as the potential advantage of a greater degree of transthyretin knockdown (89%) than has been reported with vutrisiran (81%)[77] or eplontersen (82%).[12] A phase 3 study of nex-z is ongoing (MAGNITUDE [A Phase 3 Study of NTLA-2001 in Participants With Transthyretin Amyloidosis With Cardiomyopathy (ATTR-CM)] NCT06128629),[78] with an estimated completion date of 2028. Nonetheless, the relative clinical benefit, safety, and cost of permanent modification of genomic *TTR* versus currently available transthyretin messenger ribonucleic acid silencers remain unclear.

4.3.2. Transthyretin Depleters With Monoclonal Antibodies

Current disease-modifying therapies slow or arrest the deposition of transthyretin amyloid fibrils and slow the progression of ATTR-CM but cannot reverse disease. The ability to remove transthyretin amyloid deposits, however, could restore myocardial compliance, reverse injury, and improve outcomes.[79] The development of targeted monoclonal antibodies that recognize misfolded transthyretin and facilitate degradation through activation of macrophage clearance mechanisms represents the therapeutic potential of ATTR fibril depletion agents.[80,81] One agent has completed enrollment for a phase 3 clinical trial (ALXN2220 formerly known as NI006; DepleTTR-CM [Study of ALXN2220 Versus Placebo in Adults With ATTR-CM] trial; NCT06183931[82]; estimated completion date 2027), while another is in phase 2 (coramitug, formerly known as PRX-0004; NCT05442047[83]; enrollment completed May 2025) with a phase 3 trial expected. These agents, if confirmed effective with no safety concerns, could offer a therapeutic advance for individuals later in their disease course with significant symptom burden and organ involvement, leading to "reverse remodeling" of their ATTR-CM.

4.4. Conclusions

The profound consequence of parallel advances in early recognition, accurate diagnosis, and effective management of ATTR-CM over the past decade is highlighted by the increasingly healthier trial populations in the landmark trials of amyloid therapies. Notably, in the treatment arm of the most contemporary trial, HELIOS-B, all-cause death approached the population-expected age-adjusted 3-year mortality rate.[84] Nonetheless, despite this substantial progress, many important unanswered questions remain (**Table 4**).[69]

With transthyretin-disease-modifying therapies currently approved—2 transthyretin stabilizers (acoramidis and tafamidis) and a transthyretin silencer (vutrisiran)—clinicians now have multiple options to preserve the quality of life and improve the survival of individuals living with ATTR-CM. Although no studies presently demonstrate superiority of 1 agent over another, and although data are currently insufficient to recommend routine combination therapy and a change in

TABLE 4 Important Unanswered Questions in the Management of ATTR-CM

Therapeutic Area	Question(s)	Potential Approaches to Address Emerging Questions
Initial therapy	■ Stabilizer or silencer? ■ Which agent?	■ Comparative effectiveness research using national data repositories or registries ■ Single- or multicenter collaborative research studies ■ Endpoints including progression of disease and costs in addition to mortality and cardiovascular events
Combination therapy and HF management	■ Which combination? ■ Initially or in series? ■ Choice of HF therapies	■ Randomized trials with *active controls* to determine efficacy of combination therapy compared with monotherapy ■ Comparison of combination therapy at diagnosis versus sequential initiation based on indicators of disease progression ■ Crossover studies comparing single and combination therapy ■ Randomized trials or prospective observational analyses to identify optimal combination of HF-focused therapies (including MRA and SGLT2 inhibitor)
Monitoring disease progression	■ What is the optimal combination of metrics? ■ When to reassess?	■ Determine contributions of potential indicators of therapeutic efficacy or disease progression in multiple complementary domains including circulating biomarkers (transthyretin/prealbumin,[85] troponin,[86] natriuretic peptides,[87] eGFR[87,88]), imaging biomarkers (echocardiogram GLS,[89] CMR ECV,[90] radionuclide scintigraphy[91,92]), functional markers (6MWD,[93] CPET-derived parameters[94]), quality of life (KCCQ)[95] ■ Post-hoc analysis of clinical trials, multisite registries
Prevention of disease	■ Can the development of ATTR-CM be prevented in at-risk individuals?	■ Screening in at-risk populations (SCAN-MP; NCT03812172[96]) ■ Randomized trials of stabilizers in carriers of pathogenic *TTR* variants approaching predicted age of disease onset (ACT-EARLY; NCT06563895[97]) ■ Randomized trials of disease-modifying therapies in older adults with orthopedic transthyretin amyloid identification (carpal tunnel, spinal ligament)
Cost-effectiveness	■ Costs to individuals as well as healthcare system	■ Cost-effectiveness analysis regarding efficacy vs pricing[98] ■ Analysis of access, administrative burden of the prescription process, regional variations in management

6MWD = 6-minute walk distance; ACT-EARLY = Acoramidis Transthyretin Amyloidosis Prevention Trial in the Young; ATTR-CM = transthyretin cardiac amyloidosis; CMR ECV = cardiac magnetic resonance extracellular volume; CPET = cardiopulmonary exercise stress test; ECV = extracellular volume; eGFR = estimated glomerular filtration rate; GLS = global longitudinal strain; HF = heart failure; KCCQ = Kansas City Cardiomyopathy Questionnaire; MRA = mineralocorticoid antagonist; SCAN-MP = Screening for Cardiac Amyloidosis With Nuclear Imaging in Minority Populations; SGLT2 = sodium-glucose cotransporter-2; TTR = transthyretin.

therapy to mitigate disease progression, ongoing studies may help to clarify some of the unanswered questions to promote optimal care for individuals with ATTR-CM.

PRESIDENT AND STAFF

Christopher M. Kramer, MD, FACC, President
Cathleen C. Gates, Chief Executive Officer
Richard J. Kovacs, MD, MACC, Chief Medical Officer
Brendan Mullen, Senior Executive Vice President
Joseph M. Allen, MA, Team Leader, Science & Quality
Amy Dearborn, Team Lead, Decision Science
Ashleigh M. Covington, MA, Team Lead, Decision Science Delivery
Severa Chavez, Project Manager, Decision Science Delivery
Grace D. Ronan, Senior Production and Operations Manager, Clinical Policy Publication

REFERENCES

1. Gonzalez-Lopez E, Gallego-Delgado M, Guzzo-Merello G, et al. Wild-type transthyretin amyloidosis as a cause of heart failure with preserved ejection fraction. *Eur Heart J*. 2015;36:2585–2594.

2. Castano A, Narotsky DL, Hamid N, et al. Unveiling transthyretin cardiac amyloidosis and its predictors among elderly patients with severe aortic stenosis undergoing transcatheter aortic valve replacement. *Eur Heart J*. 2017;38:2879–2887.

3. Sperry BW, Reyes BA, Ikram A, et al. Tenosynovial and cardiac amyloidosis in patients undergoing carpal tunnel release. *J Am Coll Cardiol*. 2018;72:2040–2050.

4. Westermark P, Westermark GT, Suhr OB, et al. Transthyretin-derived amyloidosis: probably a common cause of lumbar spinal stenosis. *Ups J Med Sci*. 2014;119:223–228.

5. Gillmore JD, Maurer MS, Falk RH, et al. Nonbiopsy diagnosis of cardiac transthyretin amyloidosis. *Circulation*. 2016;133:2404–2412.

6. Sperry BW, Hanna M, Shah SJ, et al. Spironolactone in patients with an echocardiographic hfpef phenotype suggestive of cardiac amyloidosis: results from TOPCAT. *JACC Heart Fail*. 2021;9:795–802.

7. Ioannou A, Massa P, Patel RK, et al. Conventional heart failure therapy in cardiac ATTR amyloidosis. *Eur Heart J*. 2023;44:2893–2907.

8. Inciardi RM, Vaduganathan M, Claggett B, et al. Application of a transthyretin amyloid cardiomyopathy score to heart failure with mildly reduced or preserved ejection fraction. *Eur J Heart Fail*. 2025;12:3012–3016.

9. Porcari A, Cappelli F, Nitsche C, et al. SGLT2 inhibitor therapy in patients with transthyretin amyloid cardiomyopathy. *J Am Coll Cardiol*. 2024;83:2411–2422.

10. Maurer MS, Schwartz JH, Gundapaneni B, et al. Tafamidis treatment for patients with transthyretin amyloid cardiomyopathy. *N Engl J Med*. 2018;379:1007–1016.

11. Gillmore JD, Judge DP, Cappelli F, et al. Efficacy and safety of acoramidis in transthyretin amyloid cardiomyopathy. *N Engl J Med*. 2024;390:132–142.

12. Fontana M, Berk JL, Gillmore JD, et al. Vutrisiran in patients with transthyretin amyloidosis with cardiomyopathy. *N Engl J Med*. 2025;392:33–44.

13. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical

Practice Guidelines. *J Am Coll Cardiol*. 2022;79:e263–e421.

14. Hicks KA, Tcheng JE, Bozkurt B, et al. 2014 ACC/AHA key data elements and definitions for cardiovascular endpoint events in clinical trials: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Data Standards (Writing Committee to Develop Cardiovascular Endpoints Data Standards). *J Am Coll Cardiol*. 2015;66:403–469.

15. Muchtar E, Gertz MA, Kyle RA, et al. A modern primer on light chain amyloidosis in 592 patients with mass spectrometry-verified typing. *Mayo Clin Proc*. 2019;94:472–483.

16. Mohammed SF, Mirzoyev SA, Edwards WD, et al. Left ventricular amyloid deposition in patients with heart failure and preserved ejection fraction. *JACC Heart Fail*. 2014;2:113–122.

17. Hahn VS, Yanek LR, Vaishnav J, et al. Endomyocardial biopsy characterization of heart failure with preserved ejection fraction and prevalence of cardiac amyloidosis. *JACC Heart Fail*. 2020;8:712–724.

18. Basdavanos A, Maurer MS, Ives L, et al. Prevalence of orthopedic manifestations in patients with cardiac amyloidosis with a focus on shoulder pathologies. *Am J Cardiol*. 2023;190:67–74.

19. Yanagisawa A, Ueda M, Sueyoshi T, et al. Amyloid deposits derived from transthyretin in the ligamentum flavum as related to lumbar spinal canal stenosis. *Mod Pathol*. 2015;28:201–207.

20. Moore ZJ, Rizkalla JM, Weiner J, et al. Transthyretin amyloidosis in spinal canal stenosis: a systematic review. *J Orthop*. 2024;53:133–139.

21. Geller HI, Singh A, Alexander KM, et al. Association between ruptured distal biceps tendon and wild-type transthyretin cardiac amyloidosis. *JAMA*. 2017;318:962–963.

22. Nakagawa M, Sekijima Y, Yazaki M, et al. Carpal tunnel syndrome: a common initial symptom of systemic wild-type ATTR (ATTRwt) amyloidosis. *Amyloid*. 2016;23:58–63.

23. Connors LH, Sam F, Skinner M, et al. Heart failure resulting from age-related cardiac amyloid disease associated with wild-type transthyretin: a prospective, observational cohort study. *Circulation*. 2016;133:282–290.

24. Kleefeld F, Scherret E, Knebel F, et al. Same same, but different? The neurological presentation of wild-type transthyretin (ATTRwt) amyloidosis. *Amyloid*. 2022;29:92–101.

25. Wajnsztajn Yungher F, Kim A, Boehme A, et al. Peripheral neuropathy symptoms in wild type transthyretin amyloidosis. *J Peripher Nerv Syst*. 2020;25:265–272.

26. Russell A, Hahn C, Chhibber S, et al. Utility of neuropathy screening for wild-type transthyretin amyloidosis patients. *Can J Neurol Sci*. 2021;48:607–615.

27. Quarta CC, Solomon SD, Uraizee I, et al. Left ventricular structure and function in transthyretin-related versus light-chain cardiac amyloidosis. *Circulation*. 2014;129:1840–1849.

28. Jiang Z, Zhuang S, Tang M, et al. Diagnostic value of the voltage-to-mass ratio in biopsy-proven cardiac amyloidosis. *Ann Noninvasive Electrocardiol*. 2024;29:e70026.

29. Dorbala S, Ando Y, Bokhari S, et al. ASNC/AHA/ASE/EANM/HFSA/ISA/SCMR/SNMMI expert consensus recommendations for multimodality imaging in cardiac amyloidosis: part 1 of 2-evidence base and standardized methods of imaging. *J Nucl Cardiol*. 2019;26:2065–2123.

30. Cuddy SAM, Chetrit M, Jankowski M, et al. Practical points for echocardiography in cardiac amyloidosis. *J Am Soc Echocardiogr*. 2022;35:A31–A40.

31. Pagourelias ED, Mirea O, Duchenne J, et al. Echo parameters for differential diagnosis in cardiac amyloidosis: a head-to-head comparison of deformation and nondeformation parameters. *Circ Cardiovasc Imaging*. 2017;10:e005588.

32. Martinez-Naharro A, Treibel TA, Abdel-Gadir A, et al. Magnetic resonance in transthyretin cardiac amyloidosis. *J Am Coll Cardiol*. 2017;70:466–477.

33. Vergaro G, Castiglione V, Aimo A, et al. N-terminal pro-B-type natriuretic peptide and high-sensitivity troponin T hold diagnostic value in cardiac amyloidosis. *Eur J Heart Fail*. 2023;25:335–346.

34. De Michieli L, AbouEzzeddine OF, Abbasi MA, et al. Biomarkers to predict abnormal technetium-99m pyrophosphate scans in patients with suspected transthyretin amyloidosis. *JACC CardioOncol*. 2025;7:70–78.

35. Davies DR, Redfield MM, Scott CG, et al. A simple score to identify increased risk of transthyretin amyloid cardiomyopathy in heart failure with preserved ejection fraction. *JAMA Cardiol*. 2022;7:1036–1044.

36. Grogan M, Lopez-Jimenez F, Cohen-Shelly M, et al. Artificial intelligence-enhanced electrocardiogram for the early detection of cardiac amyloidosis. *Mayo Clin Proc*. 2021;96:2768–2778.

37. Slivnick JA, Hawkes W, Oliveira J, et al. Cardiac amyloidosis detection from a single echocardiographic video clip: a novel artificial intelligence-based screening tool. *Eur Heart J*. 2025;46(40):4090–4101.

38. Weberling LD, Ochs A, Benovoy M, et al. Machine learning to automatically differentiate hypertrophic cardiomyopathy, cardiac light chain, and cardiac transthyretin amyloidosis: a multicenter CMR Study. *Circ Cardiovasc Imaging*. 2025;18:e017761.

39. Katzmann JA, Abraham RS, Dispenzieri A, et al. Diagnostic performance of quantitative kappa and lambda free light chain assays in clinical practice. *Clin Chem*. 2005;51:878–881.

40. Long TE, Indridason OS, Palsson R, et al. Defining new reference intervals for serum free light chains in individuals with chronic kidney disease: results of the iStopMM study. *Blood Cancer J*. 2022;12:133.

41. Kittleson MM, Ruberg FL, Ambardekar AV, et al. 2023 ACC expert consensus decision pathway on comprehensive multidisciplinary care for the patient with cardiac amyloidosis: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol*. 2023;81:1076–1126.

42. Cohen OC, Sharpley F, Gilbertson JA, et al. The value of screening biopsies in light-chain (AL) and transthyretin (ATTR) amyloidosis. *Eur J Haematol*. 2020;105:352–356.

43. Quarta CC, Gonzalez-Lopez E, Gilbertson JA, et al. Diagnostic sensitivity of abdominal fat aspiration in cardiac amyloidosis. *Eur Heart J*. 2017;38:1905–1908.

44. Fine NM, Arruda-Olson AM, Dispenzieri A, et al. Yield of noncardiac biopsy for the diagnosis of transthyretin cardiac amyloidosis. *Am J Cardiol*. 2014;113:1723–1727.

45. Gertz MA, Dispenzieri A. Systemic amyloidosis recognition, prognosis, and therapy: a systematic review. *JAMA*. 2020;324:79–89.

46. Kyle RA, Gertz MA. Primary systemic amyloidosis: clinical and laboratory features in 474 cases. *Semin Hematol*. 1995;32:45–59.

47. Hansen CT, Møller HEH, Rojek AM, et al. Combined subcutaneous fat aspirate and skin tru-cut biopsy for amyloid screening in patients with suspected systemic amyloidosis. *Molecules*. 2021;26(12):3649.

48. van Gameren I, Hazenberg BP, Bijzet J, et al. Diagnostic accuracy of subcutaneous abdominal fat tissue aspiration for detecting systemic amyloidosis and its utility in clinical practice. *Arthritis Rheum*. 2006;54:2015–2021.

49. Hutchison CA, Harding S, Hewins P, et al. Quantitative assessment of serum and urinary polyclonal free light chains in patients with chronic kidney disease. *Clin J Am Soc Nephrol*. 2008;3:1684–1690.

50. Miller EJ, Campisi R, Shah NR, et al. Radiopharmaceutical supply disruptions and the use of (99m)Tc-hydroxymethylene diphosphonate as an alternative to (99m)Tc-pyrophosphate for the diagnosis of transthyretin cardiac amyloidosis: an ASNC Information Statement. *J Nucl Cardiol*. 2022;29:2748–2760.

51. Khor YM, Cuddy SAM, Singh V, et al. (99m)Tc bone-avid tracer cardiac scintigraphy: role in noninvasive diagnosis of transthyretin cardiac amyloidosis. *Radiology*. 2023;306:e221082.

52. Damy T, Garcia-Pavia P, Hanna M, et al. Efficacy and safety of tafamidis doses in the Tafamidis in Transthyretin Cardiomyopathy Clinical Trial (ATTR-ACT) and long-term extension study. *Eur J Heart Fail*. 2021;23:277–285.

53. Lockwood PA, Le VH, O'Gorman MT, et al. The bioequivalence of tafamidis 61-mg free acid capsules and tafamidis meglumine 4 x 20-mg capsules in healthy volunteers. *Clin Pharmacol Drug Dev*. 2020;9:849–854.

54. Lee A. Acoramidis: first approval. *Drugs*. 2025;85:833–837.

55. Judge DP, Alexander KM, Cappelli F, et al. Efficacy of acoramidis on all-cause mortality and cardiovascular hospitalization in transthyretin amyloid cardiomyopathy. *J Am Coll Cardiol*. 2025;85:1003–1014.

56. Judge DP, Gillmore JD, Alexander KM, et al. Long-term efficacy and safety of acoramidis in ATTR-CM: initial report from the open-label extension of the ATTRibute-CM Trial. *Circulation*. 2025;151:601–611.

57. Brannagan TH 3rd, Berk JL, Gillmore JD, et al. Liver-directed drugs for transthyretin-mediated amyloidosis. *J Peripher Nerv Syst*. 2022;27:228–237.

58. Merative. Micromedex RED BOOK. Accessed: July 2025. https://www.merative.com/documents/micromedex-red-book

59. Bruno M, Castano A, Burton A, et al. Transthyretin amyloid cardiomyopathy in women: frequency, characteristics, and diagnostic challenges. *Heart Fail Rev*. 2021;26:35–45.

60. Tojo K, Sekijima Y, Kelly JW, et al. Diflunisal stabilizes familial amyloid polyneuropathy-associated

transthyretin variant tetramers in serum against dissociation required for amyloidogenesis. *Neurosci Res*. 2006;56:441–449.

61. Berk JL, Suhr OB, Obici L, et al. Repurposing diflunisal for familial amyloid polyneuropathy: a randomized clinical trial. *JAMA*. 2013;310:2658–2667.

62. Ambardekar AV, Fontana M, Maurer MS, et al. How to identify cardiac amyloidosis patients who might benefit from cardiac transplantation. *JACC Heart Fail*. 2023;11:115–120.

63. Adams D, Gonzalez-Duarte A, O'Riordan WD, et al. Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *N Engl J Med*. 2018;379:11–21.

64. Benson MD, Waddington-Cruz M, Berk JL, et al. Inotersen treatment for patients with hereditary transthyretin amyloidosis. *N Engl J Med*. 2018;379:22–31.

65. Adams D, Tournev IL, Taylor MS, et al. Efficacy and safety of vutrisiran for patients with hereditary transthyretin-mediated amyloidosis with polyneuropathy: a randomized clinical trial. *Amyloid*. 2023;30:1–9.

66. Witteles RM, Garcia-Pavia P, Damy T, et al. Vutrisiran improves survival and reduces cardiovascular events in ATTR amyloid cardiomyopathy: HELIOS-B. *J Am Coll Cardiol*. 2025;85:1959–1970.

67. Fontana M, Maurer MS, Gillmore JD, et al. Outpatient worsening heart failure in patients with transthyretin amyloidosis with cardiomyopathy in the HELIOS-B trial. *J Am Coll Cardiol*. 2025;85:753–761.

68. Sheikh FH, Habib G, Tang WHW, et al. Impact of vutrisiran on functional capacity and quality of life in transthyretin amyloidosis with cardiomyopathy. *J Am Coll Cardiol*. 2025;85:1943–1955.

69. Griffin JM, Grodin JL, Ruberg FL, et al. Current landscape of therapies for transthyretin amyloid cardiomyopathy. *JACC Heart Fail*. 2025;13:685–694.

70. Maurer MS. A question in the era of multiple effective therapies for transthyretin amyloid cardiomyopathy (ATTR-CM): is transthyretin biology a driver of disease progression? *J Card Fail*. 2025;31:746–747.

71. Bhuiyan T, Helmke S, Patel AR, et al. Pressure-volume relationships in patients with transthyretin (ATTR) cardiac amyloidosis secondary to V122I mutations and wild-type transthyretin: Transthyretin Cardiac Amyloid Study (TRACS). *Circ Heart Fail*. 2011;4:121–128.

72. Cheng RK, Levy WC, Vasbinder A, et al. Diuretic dose and NYHA functional class are independent predictors of mortality in patients with transthyretin cardiac amyloidosis. *JACC CardioOncol*. 2020;2:414–424.

73. Ioannou A, Cappelli F, Emdin M, et al. Stratifying disease progression in patients with cardiac ATTR amyloidosis. *J Am Coll Cardiol*. 2024;83:1276–1291.

74. Cheng RK, Vasbinder A, Levy WC, et al. Lack of association between neurohormonal blockade and survival in transthyretin cardiac amyloidosis. *J Am Heart Assoc*. 2021;10:e022859.

75. Barge-Caballero G, Barge-Caballero E, Lopez-Perez M, et al. Beta-blocker exposure and survival in patients with transthyretin amyloid cardiomyopathy. *Mayo Clin Proc*. 2022;97:261–273.

76. Lang FM, Teruya S, Weinsaft A, et al. Sodium-glucose cotransporter 2 inhibitors for transthyretin amyloid cardiomyopathy: analyses of short-term efficacy and safety. *Eur J Heart Fail*. 2024;26:938–947.

77. Fontana M, Solomon SD, Kachadourian J, et al. CRISPR-Cas9 gene editing with nexiguran ziclumeran for ATTR cardiomyopathy. *N Engl J Med*. 2024;391:2231–2241.

78. MAGNITUDE. A Phase 3 Study of NTLA-2001 in Participants With Transthyretin Amyloidosis With Cardiomyopathy (ATTR-CM). ClinicalTrials.gov ID NCT06128629. Accessed June 2025. https://clinicaltrials.gov/study/NCT06128629

79. Fontana M, Gilbertson J, Verona G, et al. Antibody-associated reversal of ATTR amyloidosis-related cardiomyopathy. *N Engl J Med*. 2023;388:2199–2201.

80. Garcia-Pavia P, Mercuri MF, Kahr PC. Antibody NI006 for cardiac transthyretin amyloid depletion. Reply. *N Engl J Med*. 2023;389:1249.

81. Suhr OB, Grogan M, Silva AMD, et al. PRX004 in variant amyloid transthyretin (ATTRv) amyloidosis: results of a phase 1, open-label, dose-escalation study. *Amyloid*. 2025;32:14–21.

82. Study of ALXN2220 Versus Placebo in Adults With ATTR-CM (DepleTTR-CM). ClinicalTrials.gov ID NCT06183931. Accessed June 2025. https://clinicaltrials.gov/study/NCT06183931

83. A Research Study to Look at How a New Medicine Called NNC6019-0001 Works and How Safe it is for People Who Have Heart Disease Due to Transthyretin (TTR) Amyloidosis, ClinicalTrials.gov ID NCT05442047. Accessed June 2025. https://clinicaltrials.gov/study/NCT05442047

84. Girard AA, Sperry BW. Contextualizing the results of HELIOS-B in the broader landscape of clinical trials for the treatment of transthyretin cardiac amyloidosis. *Heart Fail Rev*. 2025;30:69–73.

85. Maurer MS, Judge DP, Gillmore JD, et al. Early increase in serum transthyretin by acoramidis independently predicts improved survival in TTR amyloid cardiomyopathy. *J Am Coll Cardiol*. 2025;85:1911–1923.

86. Grogan M, Scott CG, Kyle RA, et al. Natural history of wild-type transthyretin cardiac amyloidosis and risk stratification using a novel staging system. *J Am Coll Cardiol*. 2016;68:1014–1020.

87. Gillmore JD, Damy T, Fontana M, et al. A new staging system for cardiac transthyretin amyloidosis. *Eur Heart J*. 2018;39:2799–2806.

88. Law S, Petrie A, Chacko L, et al. Disease progression in cardiac transthyretin amyloidosis is indicated by serial calculation of National Amyloidosis Centre transthyretin amyloidosis stage. *ESC Heart Fail*. 2020;7:3942–3949.

89. Huntjens PR, Zhang KW, Soyama Y, et al. Prognostic utility of echocardiographic atrial and ventricular strain imaging in patients with cardiac amyloidosis. *JACC Cardiovasc Imaging*. 2021;14:1508–1519.

90. Martinez-Naharro A, Kotecha T, Norrington K, et al. Native T1 and extracellular volume in transthyretin amyloidosis. *JACC Cardiovasc Imaging*. 2019;12:810–819.

91. Porcari A, Fontana M, Canepa M, et al. Clinical and prognostic implications of right ventricular uptake on bone scintigraphy in transthyretin amyloid cardiomyopathy. *Circulation*. 2024;149:1157–1168.

92. Rettl R, Wollenweber T, Duca F, et al. Monitoring tafamidis treatment with quantitative SPECT/CT in transthyretin amyloid cardiomyopathy. *Eur Heart J Cardiovasc Imaging*. 2023;24:1019–1030.

93. Ioannou A, Fumagalli C, Razvi Y, et al. Prognostic value of a 6-minute walk test in patients with transthyretin cardiac amyloidosis. *J Am Coll Cardiol*. 2024;84:43–58.

94. Patel RK, Bandera F, Venneri L, et al. Cardiopulmonary exercise testing in evaluating transthyretin amyloidosis. *JAMA Cardiol*. 2024;9:367–376.

95. Poledniczek M, Kronberger C, Willixhofer R, et al. Health-related quality of life is an independent predictor of mortality and hospitalisations in transthyretin amyloid cardiomyopathy: a prospective cohort study. *Qual Life Res*. 2024;33:2743–2753.

96. Screening for cardiac amyloidosis with nuclear imaging for minority populations (SCAN-MP). ClinicalTrials.gov ID NCT03812172. Accessed June 2025 https://clinicaltrials.gov/study/NCT03812172

97. Acoramidis transthyretin amyloidosis prevention trial in the young (act-early) study in asymptomatic carriers of a pathogenic TTR variant. ClinicalTrials.gov ID NCT06563895. Accessed June 2025. https://clinicaltrials.gov/study/NCT06563895

98. Wasfy JH, Winn AN, Touchette DR, et al. Disease modifying therapies for the treatment of transthyretin amyloid cardiomyopathy; final evidence report. *Institute for Clinical and Economic Review*. October 21, 2024. Accessed June 2025. https://icer.org/wp-content/uploads/2024/03/ICER_ATTR-CM_Final-Report_For-Publication_10212024.pdf

KEY WORDS **ACC Concise Clinical Guidance, amyloidosis, amyloid neuropathy, atrial fibrillation, cardiomyopathies, genetic testing, heart failure, heart transplantation**

JACC VOL. 87, NO. 5, 2026
FEBRUARY 10, 2026:549–565

APPENDIX 1. AUTHOR RELATIONSHIPS WITH INDUSTRY AND OTHER ENTITIES (RELEVANT)—TRANSTHYRETIN CARDIAC AMYLOIDOSIS EVALUATION AND MANAGEMENT: 2025 ACC CONCISE CLINICAL GUIDANCE

Committee Member	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Michelle M. Kittleson, *Chair*	Smidt Heart Institute, Cedars-Sinai Medical Center—Professor of Medicine	None	None	None	None	None	None
Amrut V. Ambardekar	University of Colorado — Professor of Medicine-Cardiology	None	None	None	None	None	None
Richard K. Cheng	University of Washington—Professor of Cardiology	None	None	None	None	None	None
Jan M. Griffin	Medical University of South Carolina—Associate Professor of Medicine	■ AstraZeneca Pharmaceuticals ■ BridgeBio ■ Pfizer Inc	■ BridgeBio	None	■ Pfizer*	■ BridgeBio* ■ Scripps*	None
Mathew S. Maurer	Columbia University Irving Medical Center—Professor of Medicine	■ AstraZeneca Pharmaceuticals* ■ Attralus ■ BridgeBio ■ Eidos, Inc ■ Ionis ■ Novo Nordisk Inc*	None	None	■ Alnylam Inc ■ Intellia (DSMB)*	None	None
Jose Nativi-Nicolau	Mayo Clinic Florida—Transplant Cardiologist	None	None	None	None	None	None
Frederick L. Ruberg	Boston University—Professor of Medicine	■ AstraZeneca ■ eMyosound	None	None	■ Akcea Therapeutics* ■ Alnylam Pharmaceuticals* ■ Anumana* ■ National Institutes of Health* ■ Pfizer, Inc* ■ TriNetX	None	None

This table represents all relationships with industry and other entities that were reported by authors, including those not deemed to be relevant to this document, at the time this document was under development. The table does not necessarily reflect relationships with industry at the time of publication. A person is deemed to have a significant interest in a business if the interest represents ownership of ≥5% of the voting stock or share of the business entity, or ownership of ≥$5,000 of the fair market value of the business entity, or if funds received by the person from the business entity exceed 5% of the person's gross income for the previous year. Relationships that exist with no financial benefit are also included for the purpose of transparency. Relationships in this table are modest unless otherwise noted. Please refer to https://www.acc.org/guidelines/about-guidelines-and-clinical-documents/relationships-with-industry-policy for definitions of disclosure categories or additional information about the ACC Disclosure Policy for Writing Committees.
*Significant relationship.

ACC = American College of Cardiology; DSMB = data and safety monitoring board.

APPENDIX 2. PEER REVIEWER RELATIONSHIPS WITH INDUSTRY AND OTHER ENTITIES (COMPREHENSIVE)—TRANSTHYRETIN CARDIAC AMYLOIDOSIS EVALUATION AND MANAGEMENT: 2025 ACC CONCISE CLINICAL GUIDANCE

Peer Reviewer	Representation	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness
Nadia Fida	Official Reviewer—ACC Heart Failure & Transplant Council	Houston Methodist Baytown—Advanced Heart Failure and Transplant Cardiologist; Medical Director of Cardiology	None	None	None	■ CVRx	None	None
Robert C Hendel	Official Reviewer—Solution Set Oversight Committee	Tulane School of Medicine—Professor of Medicine and Radiology	None	None	None	None	None	None
Chayakrit Krittanawong	Official Reviewer—Solution Set Oversight Committee	New York University Langone Health—Cardiologist	None	None	None	None	None	None
Senthil Selvaraj	Content Reviewer—ACC Expert	Duke University—Assistant Professor of Medicine	■ AstraZeneca Pharmaceuticals ■ BridgeBio ■ PeerView	None	None	None	■ Heart Failure Society of America	None
Brett Wesley Sperry	Content Reviewer—ACC Expert	Saint Luke's Mid America Heart Institute—Cardiologist	■ Alnylam ■ AstraZeneca Pharmaceuticals ■ Eidos/BridgeBio	■ Alnylam ■ Pfizer Inc*	None	None	■ ATTRibute-CM, Eidos† ■ CARDIO-TTRansform, Ionis† ■ DepleTTR, Alexion†	None

This table represents all relationships with industry and other entities that were reported by peer reviewers, including those not deemed to be relevant to this document, at the time this document was under development. The table does not necessarily reflect relationships with industry at the time of publication. A person is deemed to have a significant interest in a business if the interest represents ownership of ≥5% of the voting stock or share of the business entity, or ownership of ≥$5,000 of the fair market value of the business entity; or if funds received by the person from the business entity exceed 5% of the person's gross income for the previous year. Relationships that exist with no financial benefit are also included for the purpose of transparency. Relationships in this table are modest unless otherwise noted. Please refer to https://www.acc.org/guidelines/about-guidelines-and-clinical-documents/relationships-with-industry-policy for definitions of disclosure categories or additional information about the ACC Disclosure Policy for Writing Committees.

*Significant relationship.

†Relationship with this company is limited to enrolling patients in clinical trials. This disclosure was entered under the Clinical Trial Enroller category in the ACC's disclosure system. To appear in this category, the peer reviewer acknowledges that there is no direct or institutional relationship with the trial sponsor as defined in the ACC Disclosure Policy for Writing Committees.

ACC = American College of Cardiology; ATTRibute-CM = Acoramidis in Transthyretin Cardiomyopathy Clinical Trial; CARDIO-TTRansform = Evaluate the Efficacy and Safety of Eplontersen in Participants With Transthyretin-Mediated Amyloid Cardiomyopathy Clinical Trial; DepleTTR-CM = Study of ALXN2220 Versus Placebo in Adults With ATTR-CM Clinical Trial.